

Assessment of the effects of nickel on benthic macroinvertebrates in the field

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Abstract A field-based evaluation of the biological effects of potential nickel (Ni) exposures was conducted using monitoring data for benthic macroinvertebrates and water chemistry parameters for streams in England and Wales. Observed benthic community metrics were compared to expected community metrics under reference conditions using RIVPACS III+ software. In order to evaluate relationships between Ni concentrations and benthic community metrics, bioavailable Ni concentrations were also calculated for each site. A limiting effect from Ni on the 90th percentile of the maximum achievable ecological quality was derived at “bioavailable Ni” exposures of $10.3 \mu\text{g l}^{-1}$. As snails have been identified as particularly sensitive to nickel exposure, snail abundance in the field in response to nickel exposure, relative to reference conditions, was also analysed. A “low effects” threshold for snail abundance based on an average of spring and autumn data was derived as $3.9 \mu\text{g l}^{-1}$ bioavailable Ni. There was no apparent effect of Ni exposure on the abundance of Ephemeroptera (mayflies), Plecoptera (stoneflies) or Trichoptera (caddisflies) when expressed relative to a reference condition within the range of “bioavailable Ni” exposures observed within the dataset. Nickel exposure concentrations co-vary with the concentrations of other stressors in the dataset, and high concentrations of Ni are also associated with elevated concentrations of other contaminants.

Keywords Nickel · Invertebrate · Snail · Bioavailability

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Introduction

Ecotoxicity testing has revealed that *Lymnaea stagnalis*, a gastropod mollusc, is particularly sensitive to nickel (Ni) exposure (Schlekat et al. 2010). This pond snail is reported to be common in the UK (Environment Agency 1997) and parts of Europe, and the family Lymnaeidae are included in both the Biological Monitoring Working Party (BMWP) scoring system (Biological Monitoring Working Party, 1978) and River Invertebrate Prediction and Classification System (RIVPACS) software, which are tools that the UK environmental regulators use routinely for environmental quality assessment based on ecological monitoring in the field.

RIVPACS methodologies are also used in other EU member states (such as Spain, Sweden and the Czech Republic) and underpin the “reference-based” classification of waterbody ecological status required under the EU Water Framework Directive (WFD). Failures of the current hardness-banded statutory Ni Environmental Quality Standard (EQS) are relatively uncommon in the UK. EQS values vary between 50 and $200 \mu\text{g l}^{-1}$ (dissolved Ni) depending on hardness, which is generally above all but the most extreme nickel exposures observed in the UK. An interim EQS of $20 \mu\text{g l}^{-1}$ (dissolved Ni) has been adopted under the WFD (European Union 2008), and a revised EQS of $4 \mu\text{g l}^{-1}$ (bioavailable Ni) has recently been proposed by the European Commission (European Commission 2012) under the WFD review of priority substances.

Macroinvertebrate fauna in moving freshwaters (rivers and streams) are regularly monitored by the Environment Agency in England and Wales to determine their ecological quality. Environmental quality can be summarised using a series of biotic metrics (i.e. number of BMWP scoring taxa present [N-TAXA] and average BMWP score per taxon found [ASPT]). Different waterbodies will naturally support different macroinvertebrate communities due to differences

in habitat (e.g. geography, climate, flow, physicochemistry and substrate). The values of biotic indices derived for different waterbodies will vary because of these factors, even if their environmental quality is the same. Therefore, biotic metrics themselves cannot be directly used to compare the water quality at different sites, unless the sites are very similar in habitat.

In order to overcome this problem, predictive tools such as RIVPACS (Clarke et al. 2003) can be used to express the expected biotic metrics for specific sites under reference conditions based on site-specific abiotic characterisation information such as altitude, gradient, substrate type, flow, alkalinity and distance from source of the sampling site and extrapolation to a network of “reference sites”. This allows biotic metrics for a site to be expressed as Environmental Quality Indices (EQI). An EQI is the observed biotic metric at a site (e.g. ASPT score) divided by the expected value of that biotic index for the same site when it is in an unimpacted state. The EQI effectively removes the effect of natural differences that occur in biotic metrics at different sites and allows greater focus on sites where environmental quality may be degraded.

Macroinvertebrate communities are comprised of organisms that exhibit diverse ecologies and life histories. They can be exposed to contaminants through both the water column and sediments (suspended or settled) as well as via diet. The relative importance of each pathway for a particular chemical depends on a combination of the fate and behaviour of an individual contaminant and the ecology of the individual species. Some organisms are more exposed to sediments, i.e. they are obligate infauna (e.g. oligochaetes). Others are epifauna (e.g. Plecoptera) that are exposed predominantly through the water column and their food. The overall effect of a contaminant can be observed in the response of the community as a whole. The rivers and streams which are sampled for benthic macroinvertebrate quality in the UK are typically relatively small, steep and fast-flowing. They do not tend to contain large quantities of fine sediments, and so exposure via the water phase is likely to be the main exposure route at the monitoring sites.

Recent UK studies have suggested that comparisons based on EPT taxa (Ephemeroptera [mayflies], Plecoptera [stoneflies] and Trichoptera [caddisflies]) may not be ideal as these appear to be relatively insensitive to the effects of some metals (Environment Agency 2008). This is contrary to the apparent sensitivity to metals of these groups of species reported in North American studies (e.g. Carlisle and Clements 1999). In the UK, the assessment of the effect of Ni on community structure is less clear because zinc was the major contaminant at the majority of sites in the Environment Agency (2008) study at which Ni was also found. In addition, Crane et al. (2007) also found that EPT taxa may be relatively sensitive to Ni exposure.

The two most sensitive species in the ecotoxicity database for Ni are both invertebrates (*L. stagnalis* and *Ceriodaphnia*

dubia; DEPA 2010). In the absence of any information on differences in sensitivity between lentic and lotic systems, this suggests that invertebrates are the most appropriate trophic level at which to look for possible effects due to Ni in the field. Routine ecological monitoring of benthic macroinvertebrates, such as that used for ecological quality classification under the WFD, represents an attractive potential data source for performing an assessment of the field response of benthic macroinvertebrates to nickel exposure, and has been used for the derivation of other quality standards (UKTAG 2006). This is because the techniques for ecological monitoring across Europe under the WFD are standardised, intercalibrated and normalised to reference condition (European Commission WFD website, http://ec.europa.eu/environment/water/water-framework/objectives/status_en.htm#_More_information; UK WFD TAG website, <http://www.wfduk.org/reference/summaries-biological-status-methods>).

The presence of a biological species, or group of species, known to be sensitive to a particular type of pollutant provides strong evidence that the chemical is either not present or not bioavailable (Clements and Newman 2002). A comparison of field observations (O) against expectations from RIVPACS (E) allows ecological impacts to be assessed as the ratio of O/E (or as an EQI for whole communities). In cases where a particular taxon, or group of taxa, would not be expected to be present at a site due to the habitat being unsuitable, then this would not be interpreted as a potential effect due to Ni. This approach requires taxon-specific RIVPACS predictions and monitoring information to be available rather than just summaries of EQI N-TAXA and ASPT.

The objectives of the present study were to:

1. Make predictions of Ni bioavailability at ecological monitoring sites based on local water chemistry
2. Express the quality of the macroinvertebrate community observed at monitoring sites relative to reference conditions for both the whole community metrics such as ASPT and N-TAXA and specific scoring taxa, i.e. Lymnaeidae
3. Compare O/E ratios for the whole community and snail taxa against Ni exposure
4. Determine a level of bioavailable Ni exposure which is consistent with attaining good ecological status under the Water Framework Directive
5. Determine a level of bioavailable Ni exposure which is consistent with there being no observable response on sensitive subsets of the benthic macroinvertebrate community

Methods

Comprehensive ecological survey information for benthic macroinvertebrates and associated water chemistry parameters

for river and stream sampling sites in England, Wales and Scotland was provided by the Environment Agency. This dataset was prepared by Staffordshire University's Centre for Intelligent Environmental Systems (CIES). These biology and chemistry data had been matched spatially and temporally by CIES. A simplified form of the nickel biotic ligand model (Nickel Screening Tool) has been developed to assist in the assessment of nickel bioavailability in freshwaters (<http://www.nickelinstitute.org/en/NiPERA/EnvironmentalScience/FS4-BioavailabilityAquaticCompartment.aspx>). The Nickel Screening Tool was used to calculate bioavailable Ni concentrations for each sample. Multiple samples for the same site were treated separately, and chemical monitoring data were expressed as the median concentration for the 12 months prior to the relevant ecological sample.

Ecological (benthic macroinvertebrate) monitoring data and quality assessment

Ecological monitoring data, as benthic macroinvertebrates, were obtained using the RIVPACS semi-quantitative sampling approach (Murray-Bligh et al. 1997, Environment Agency 1999, Wright 2000) which includes sampling of benthic macroinvertebrates from all major habitats at a site during a 3-min “kick” sampling period. In addition, a further 1-min hand search is conducted for taxa likely to be missed in the initial sample (e.g. those adhering to large stones and bedrock). In this manner, the full range of benthic macroinvertebrates at a site is incorporated into the sample. Taxa are identified to family level, and their presence and abundance are recorded. The abundance of each taxon is reported either as the number of individuals present or as a log abundance score (1: one to 9 individuals, 2: ten to 99 individuals, 3: 100–999 individuals, 4: 1,000–9,999 individuals, 5: 10,000+ individuals). Subsequent statistical processing of abundance data in RIVPACS is based on the log abundance score only.

The assessment of macroinvertebrate community and individual taxa quality at individual sites was carried out using RIVPACS III+ software (Clarke et al. 2003), which allows the comparison of observed quality at a site to the expected quality based on reference (unimpacted) conditions. Macroinvertebrate community quality at sites is expressed as summary metrics based on the BMWP scoring system (Biological Monitoring Working Party, 1978). Within the BMWP system, each macroinvertebrate family (taxa) is assigned a specific score, based on its sensitivity to organic (sanitary) pollution. Low-scoring taxa are more tolerant of organic pollution. BMWP metrics are assigned based on the presence of scoring taxa rather than their abundance.

ASPT relates solely to organic pollution (Murray-Bligh et al. 1997). It is insensitive to toxic pollution such as

acidification and metalliferous discharges. N-TAXA is sensitive to toxic pollution, as well as organic pollution. N-TAXA will also respond to other environmental disturbances including the physical degradation of habitats, for instance, by siltation or channelization.

The BMWP score responds to both organic pollution and toxic pollution. However, it provides no additional information to that provided by ASPT and N-TAXA and is therefore not used for water quality classification in the UK (Murray-Bligh et al. 1997).

When ASPT and N-TAXA summary metrics are expressed as an observed (O)/expected (E) ratio based on predictions to reference (unimpacted) conditions at a site, they are referred to as EQI. EQI values <1 indicate deviation of the benthic macroinvertebrate fauna at a site from that expected under reference (unimpacted) conditions. EQI values at sites can exceed 1. EQI ASPT and EQI N-TAXA summary statistics can be calculated from monitoring data pooled across several seasons within a single year (i.e. spring and autumn) or from data collected on a single sampling occasion. EQIs based on ASPT and N-TAXA can be interpreted independently; however, additional information on the factors contributing to poor environmental quality can be obtained by interpreting them together (Murray-Bligh et al. 1997). Where EQI ASPT is low but not EQI N-TAXA, organic pollution is likely to be present on its own. Organic pollution is also likely when both EQI ASPT and EQI N-TAXA are low, though toxic pollution or habitat degradation could also be responsible. Where EQI ASPT is high but EQI N-TAXA is low, toxic pollution or habitat degradation is likely to be the cause, and organic pollution is considered unlikely (Murray-Bligh et al. 1997).

EQI thresholds of 0.9 and 0.7 are used for ASPT and N-TAXA, respectively, to define the boundary between “good” and “less than good” status during benthic macroinvertebrate quality classification in the UK for the WFD (UK WFD TAG website, <http://www.wfduk.org/reference/summaries-biological-status-methods>). A waterbody that fails to achieve good status is a candidate for regulatory intervention. Benthic macroinvertebrate quality classification in the UK is performed using the River Invertebrate Classification Tool (RICT; <http://rict.org.uk/>), which is an updated version of RIVPACS III+. However, as RICT is unable to perform taxa-specific predictions of abundance or presence/absence, it was not suitable for the analysis required and RIVPACS III+ was used preferentially. EQI predictions made using RIVPACS III+ and RICT are considered to be sufficiently similar such that its use does not compromise the objectives of the analysis in terms of WFD classification.

Despite the availability of reference-based classification tools and summary metrics such as those described earlier, it is possible that a decline in the abundance of a particularly

sensitive group of taxa may occur in cases where the quality of the overall community has not been compromised, to the extent that can be observed using the available assessment tools. There is some concern that snails may be particularly sensitive to several metals, including Ni, predominantly because of the reported sensitivity of *L. stagnalis* in laboratory tests (Schlekat et al. 2010, Brix et al. 2012). To assess this, O/E ratios for selected groups of individual taxa were calculated using RIVPACS III+ alongside the conventional EQI ASPT and EQI N-TAXA summary metrics. These O/E scores were calculated as the sum of the observed log abundances of each taxon, divided by the sum of the expected log abundances in each sample, i.e. relative to the predicted abundance under reference conditions. This analysis was applied to the following groups of taxa: snails, Ephemeroptera, Plecoptera and Tricoptera. As deviation from reference conditions is only considered to occur below an O/E value of 1, an O/E threshold of 0.9 was adopted as a threshold for biological significance in this analysis. This threshold is considered to be conceptually comparable to the use of an EC10 from a laboratory toxicity test as equivalent to a “no observed effect concentration” and is similar to the thresholds for EQI ASPT and EQS N-TAXA adopted for surface water classification on a whole community basis. This threshold was considered appropriate as a consequence of the inherent uncertainty of the O/E assessment which is comprised of elements related to the suitability of “reference” sites in the RIVPACS database to individual sites and differences in the expression of abundance data in RIVPACS, i.e. RIVPACS uses a discrete log abundance scale as observed data but reports expected data on a continuous scale. As a result, high O/E values can be calculated for sites, which should be considered an artefact. For this reason, a decline from the maximum achievable ecological quality was not considered to be relevant.

Chemical monitoring data

The chemical monitoring dataset is predominantly focused on chemical parameters which are relevant to potential impacts from sewage discharges and trace metals. Metal concentrations are reported as both total and dissolved concentrations in many cases. The dataset does not include any information on dissolved organic carbon (DOC) concentrations in the watercourses. Concentrations of DOC (mg l^{-1}) were estimated from dissolved iron concentrations (mg l^{-1}) according to Eq. 1, following our observation of a correlation between these two parameters in several UK datasets (Peters et al. 2011). The relationship used is based only on data for 35 sites reported in the Environment Agency (2008) report. The r^2 for the regression was 0.825 (adjusted r^2 0.820), with a p value <0.001 . Figure 1 shows this

regression plotted along with additional data ($n=372$) from UK monitoring programmes undertaken by the Environment Agency and Scottish Environment Protection Agency in order to confirm the validity of the relationship in a broader range of waters.

$$\text{DOC} = 1 + 21 \times \text{Fe} \quad (1)$$

An upper limit of 10 mg l^{-1} DOC was applied as a precautionary measure in order to prevent iron-impacted sites from resulting in very high estimated DOC concentrations, although predictions do appear to be robust even at relatively high dissolved iron concentrations (i.e. above 0.5 mg l^{-1} dissolved iron).

Nickel exposures

Chronic biotic ligand models (BLM) are currently available for describing Ni toxicity to algae, plants, invertebrates and fish (De Schamphelaere et al. 2006, Deleebeeck et al. 2007, Deleebeeck et al. 2008, Deleebeeck et al. 2009). Based on a cross-species validation (Schlekat et al. 2010), these models are used to calculate the specific sensitivities of all of the organisms in the aquatic effects database for specific sets of conditions, allowing a site-specific species sensitivity distribution (SSD) to be calculated. The SSD is summarised in terms of the Ni concentration which would be toxic to 5 % of the species: known as the HC5. HC5 values therefore express the sensitivity of the ecosystem to Ni under a particular set of abiotic conditions. A simplified Nickel Bioavailability Screening Tool has been developed, which

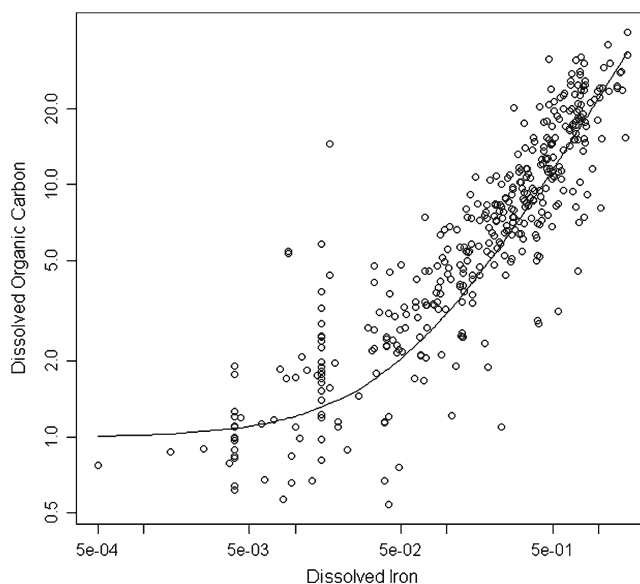


Fig. 1 DOC (mg l^{-1}) and dissolved iron (mg l^{-1}) monitoring data from the UK. The solid line indicates the relationship used to estimate DOC concentrations

simplifies the approach to calculate a site-specific HC5, although with slightly lower accuracy than the BLM.

The sensitivity of sites to Ni was predicted on a sample-specific basis using the Nickel Bioavailability Screening Tool. Sample-specific annual median data for pH, DOC (estimated from dissolved iron concentrations) and Ca were used to calculate a “bioavailability factor” (BioF value) in order to convert the dissolved Ni concentrations into bioavailable Ni concentrations. The BioF value is calculated as the reference HC5 divided by the site-specific HC5. The reference HC5 is set as the HC5 for Ni under conditions of extremely high bioavailability, which occur infrequently within Europe.

This information was used to express Ni exposures as bioavailable Ni concentrations (in $\mu\text{g l}^{-1}$), which are calculated as the product of the dissolved Ni concentration and the BioF value. This process takes account of differences in bioavailability amongst the sites.

Influence of DOC estimates on calculated nickel bioavailability

The influence of using the estimation method for DOC concentrations from dissolved iron concentrations was examined by applying the Ni bioavailability screening tool to an independent dataset with matched measured data for both DOC and dissolved iron, as well as the other required parameters (pH and Ca). Monitoring data collected from 112 sites in England and Wales were provided by the Environment Agency. These data were used to calculate the Ni HC5 using both measured and estimated DOC data in order to assess the influence of estimating DOC concentrations from dissolved iron concentrations.

Statistical analysis

Matched data for “bioavailable Ni” and ecological metrics, expressed either as EQI values (for the whole community) or as O/E values (for subsections of the community), were analysed by quantile regression in order to identify a bioavailable Ni concentration which is protective of the benthic invertebrate community or parts of that community. The quantile regression approach can be used to derive a limiting function, when applied at high quantiles, and is particularly useful in cases where it is not possible to remove all of the potentially confounding factors (predominantly impacts at sites caused by other stressors). Quantile regression has been applied to the concept of limiting factors as constraints on organisms previously (Scharf et al. 1998, Cade et al. 1999, Cade and Noon 2003). The approach taken in this study was similar to that applied in previous assessments of field data (Pacheco et al. 2005, Linton et al. 2007, Crane et al. 2007). An advantage of statistical techniques such as

quantile regression is that they do not require the datasets to be screened for potentially interfering pressures such as habitat alteration, flow regimes, invasive species or other chemical contaminants.

The principle of the limiting function approach is that, at low exposure concentrations, the maximum achievable ecological quality will not be limited by Ni, but at elevated exposure concentrations the maximum achievable ecological quality will be limited by Ni, regardless of the concentrations of other contaminants. In cases where other contaminants or pressures result in reduced ecological quality, the quantile regression approach effectively ignores these in the analysis. There is an assumption that the maximum achievable ecological quality will be achieved at sites where Ni is the only significant potential limiting factor. Other contaminants which co-vary in their concentrations with Ni may result in the limiting functions derived being lower than they would be for Ni-only exposures.

Quantile regression was applied to the datasets using a log linear model ($\log(\text{response}) = a + b \cdot (\text{exposure})$) using R (Version 2.9.0; R Core Team, 2012). The quantile regression model was applied to the 90th quantile of the datasets. Confidence intervals were calculated by bootstrapping using 2,000 resamples. A decline, as indicated by quantile regression, was considered to be significant if the slope parameter was both negative and significantly different from 0 ($p < 0.05$). The principle of the quantile regression approach to derive a limiting function is shown in Fig. 2. Data with unimpacted responses at low exposures effectively serve as controls, and impacted data at low exposures are effectively not considered in the analysis. A decline in the maximum response with increasing exposure is assumed to be due to the contaminant of interest. No methods exist for the aggregation of data from individual sampling seasons when assessing subsections of the community, although such methods do exist for the summary metrics used to describe the whole benthic invertebrate

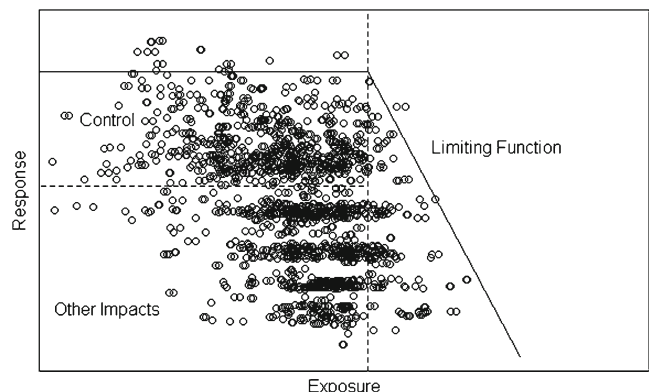


Fig. 2 Illustration of the interpretation of combined chemical and ecological field data

community. Therefore, separate assessments were performed for spring and autumn samples.

Where applicable, as co-variance of other contaminants with nickel can result in overly precautionary thresholds, an analysis of co-variance using R was undertaken to understand the statistical significance and strength of any co-variance between chemicals used in the thresholds analysis (Version 2.9.0; R Core Team, 2012). As all of the potential contaminants are not analysed at all of the sites, any correlation was assessed using each site where both parameters (i.e. Ni and the parameter being assessed) had been included in the analysis. This maximises the number of observations which can be included in the analysis. Spearman's rank correlation coefficients were used as this method is non-parametric and therefore does not depend on the distribution of the data.

Results

Estimation of DOC concentrations

The effect of the estimation of DOC concentrations on the calculation of HC5 values for 112 UK sites is shown in Fig. 3. The majority (>98 %) of the HC5 predictions using estimated DOC concentrations were within a factor of 2 of the result calculated using measured DOC data for the same samples and do not show any tendency for the estimated DOC concentrations to result in a clear bias of the HC5 predictions, either towards higher or lower values. The average error in the HC5 estimates for this dataset was $2.4 \mu\text{g l}^{-1}$, or 28 % if

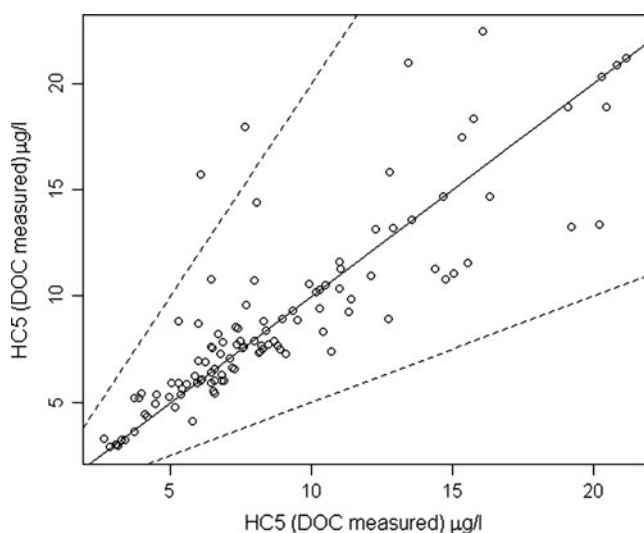


Fig. 3 Effect of estimating DOC concentrations in the calculated HC5 ($\mu\text{g l}^{-1}$ dissolved Ni) for 112 UK sites. The *solid line* indicates a 1:1 relationship between the HC5 calculated using both measured and estimated DOC concentrations, and the *dashed lines* indicate a factor of two deviations from this result

expressed as a relative error, of the HC5 value calculated using measured DOC concentrations.

This indicates that the estimation method used for calculating DOC concentrations was reliable, although there was some loss in the accuracy of bioavailability predictions.

Whole community response to nickel

No significant decline was observed in ecological quality when expressed as EQI ASPT over the range of nickel exposures. As discussed, ASPT is sensitive to pressures related to sanitary contamination. However, a significant decline was observed in ecological quality when expressed as EQI N-TAXA, which is sensitive to toxic pollution. The fitted model for the 90th percentile of the EQI N-TAXA was normalised to a value of 1 at low exposures, and a threshold concentration above which 90 % of sites would not be expected to achieve good ecological status, as defined under the WFD, was derived as $10.3 \mu\text{g l}^{-1}$ bioavailable nickel (95 % confidence interval 9.0 to $18.2 \mu\text{g l}^{-1}$, $p < 0.001$; see Fig. 4). Dissolved Ni concentrations reported in the matched dataset ranged from 0.5 to $166 \mu\text{g l}^{-1}$, and bioavailable Ni concentrations ranged from 0.1 to $21 \mu\text{g l}^{-1}$. The 95th and 99th percentile bioavailable Ni concentrations were 3.4 and $7.8 \mu\text{g l}^{-1}$, respectively. This analyses suggests that high Ni concentrations are not compatible with achieving good ecological status as determined using benthic macroinvertebrates.

Taxa-specific response to nickel

The results of the analyses of specific groups of taxa are shown in Table 1. Of the four groups of taxa assessed, only the snails

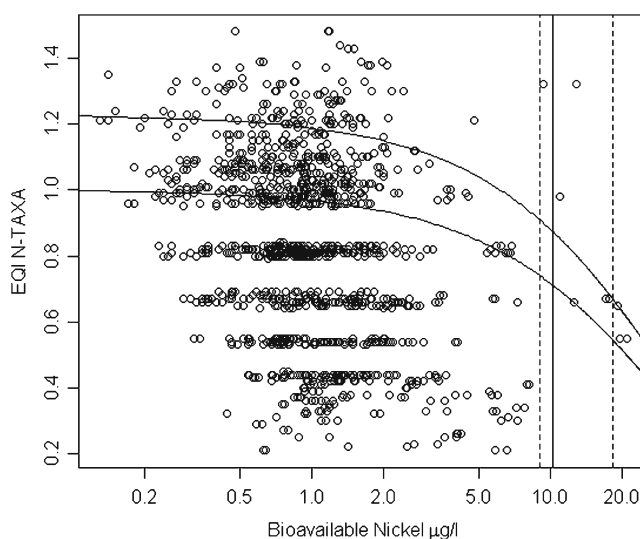


Fig. 4 EQI N-taxa as a function of bioavailable nickel concentrations (mg l^{-1}). The fitted quantile regression model is indicated by the *solid curve*, and the threshold for good ecological status by the *solid vertical line*, and the *dashed vertical lines* indicate the 95 % confidence interval

showed a significant decline in both the spring ($p=0.001$; Fig. 5) and autumn ($p<0.001$; Fig. 6) analyses by quantile regression. The values for the O/E threshold of 0.9 were $4.19 \mu\text{g l}^{-1}$ (95 % confidence interval 3.68 to $5.06 \mu\text{g l}^{-1}$) and $3.65 \mu\text{g l}^{-1}$ (95 % confidence interval 3.06 to $3.95 \mu\text{g l}^{-1}$) in the spring and autumn, respectively.

Ephemeroptera, Plecoptera and Tricoptera were not significantly affected by Ni within the range of exposures found in the field (Fig. 7). Tricoptera showed a significant decline in the autumn ($p=0.016$), but not in the spring ($p=0.167$). The importance of a significant decline being observed only in a single season is unclear, although in the case of Tricoptera any thresholds derived would be higher than those for snails in the same season. The thresholds derived for snails in the spring and autumn are shown in Table 1, along with lower and upper 95 % confidence limits. Snail abundance appears to be slightly more sensitive to Ni in autumn than in spring, although the difference is small.

Discussion

The normalisation of toxicity data for species in the ecotoxicity dataset was performed using a BLM developed for the same trophic level. Two BLMs were developed for the invertebrates *Daphnia magna* and *Ceriodaphnia dubia*, and when applying these toxicity data to other invertebrates, it is not clear which of these models best describes the response. In the nickel screening tool, the most conservative model was used, although a recent study (Heijerick and Vercaigne 2011) suggests that this could potentially result in HC5 predictions which are overprotective, especially above pH 7. Ninety percent of the data in the field dataset analysed here were above pH 7, and 50 % were above pH 7.5. The effect of precautionary predictions of nickel HC5 on this analysis would be for the calculated bioavailable nickel concentrations to be expressed as slightly higher than their true values (Table 2). This could potentially result in the derived thresholds being higher than their true values. As the most appropriate normalisation model is not known for the majority of the BMWP scoring families, the use of a

more stringent invertebrate normalisation model is considered to be the most appropriate approach for the analyses in the present study.

The threshold for bioavailable nickel of $10.3 \mu\text{g l}^{-1}$ derived from the analysis of the response of the EQI N-TAXA metric has a direct link to the classification of surface waters according to their ecological status under the WFD and suggests that levels of bioavailable Ni which are greater than twice the EQS of $4 \mu\text{g l}^{-1}$ (bioavailable Ni) currently proposed by the EU would not compromise the ability of benthic macroinvertebrate communities to achieve good ecological status. However, given that some deterioration from “reference conditions” is consistent with the normative definition of good ecological status given in Article V of the WFD, it remains a possibility that there could be other, more subtle, effects on particularly sensitive subsets of the community in response to nickel below the $10.3 \mu\text{g l}^{-1}$ bioavailable nickel threshold.

The fact that no significant decline was observed for EQI ASPT in response to increasing “bioavailable Ni” concentrations, but a significant decline was observed for EQI N-Taxa, suggests that sites which receive the highest “bioavailable Ni” exposures are subject to toxic pressures rather than pressures related to organic pollution (Murray-Bligh et al. 1997).

Thresholds for nickel-sensitive taxa

Relatively low thresholds were derived by Crane et al. (2007) for the effects of dissolved Ni on the abundance of Ephemeroptera ($<1 \mu\text{g l}^{-1}$) and the abundance of Ephemeroptera, Plecoptera and Tricoptera (approximately $2 \mu\text{g l}^{-1}$) using a different statistical analysis (piecewise regression). Thresholds were derived by Crane et al. (2007) by 99th percentile quantile regression. However, the study did not take account of the bioavailability of Ni when assessing exposure, and the ecological metrics were not expressed relative to a reference condition (i.e. the response in the metrics may have been through natural variation in habitat rather than in response to nickel). These two factors, along with the different statistical analyses performed, are likely to explain the difference between

Table 1 Summary of analysis of the effect of Ni on the abundance of groups of invertebrate taxa in the field, including O/E 0.9 threshold and lower (LCL) and upper (UCL) 95 % confidence limits where calculable

Season	Taxa	Slope	P	Interpretation	O/E 0.9	LCL	UCL
Spring	Snails	-0.2036	0.0008	Significant	4.19	3.68	5.06
Spring	Ephemeroptera	0.0207	0.7774	No decline	NA	NA	NA
Spring	Plecoptera	-0.0719	0.6896	Not significant	NA	NA	NA
Spring	Tricoptera	-0.1169	0.1674	Not significant	NA	NA	NA
Autumn	Snails	-0.2267	0.0000	Significant	3.65	3.06	3.95
Autumn	Ephemeroptera	-0.0944	0.0746	Not significant	NA	NA	NA
Autumn	Plecoptera	-0.0730	0.5398	Not significant	NA	NA	NA
Autumn	Tricoptera	-0.0803	0.0164	Significant	7.89	5.06	14.92

NA no threshold or confidence interval could be calculated

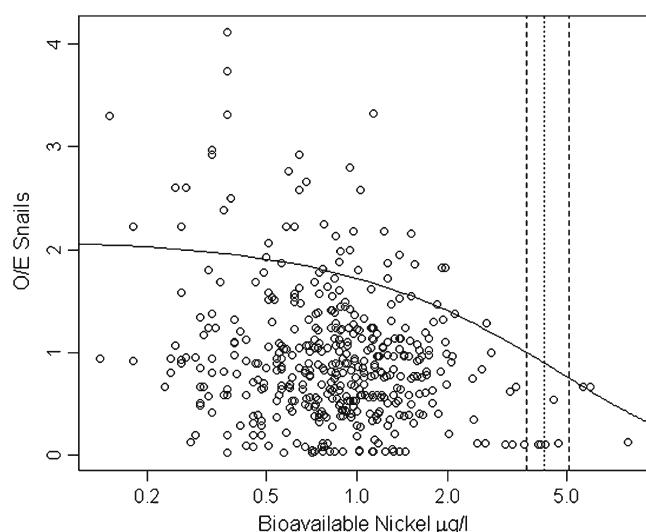


Fig. 5 O/E for snail abundance as a function of bioavailable nickel concentrations in spring. The *solid line* indicates the fitted quantile regression model, and the *vertical dashed line* indicates the value at which O/E=0.9

the findings of Crane et al. (2007) and this study and highlight the importance of accounting for both bioavailability and ecological variability when undertaking an assessment of this type.

An overall concentration of $3.9 \mu\text{g l}^{-1}$ “bioavailable Ni” is considered to be consistent with no observable decline in snail abundance from reference conditions. This value is derived as the arithmetic mean of the O/E 0.9 thresholds derived for spring and autumn (the geometric mean is also $3.9 \mu\text{g l}^{-1}$).

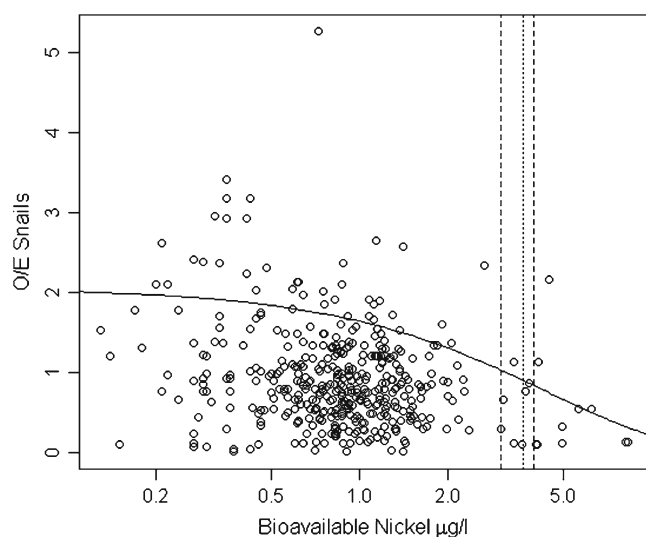


Fig. 6 O/E for snail abundance as a function of bioavailable nickel concentrations in autumn. The *solid line* indicates the fitted quantile regression model, and the *vertical dashed line* indicates the value at which O/E=0.9

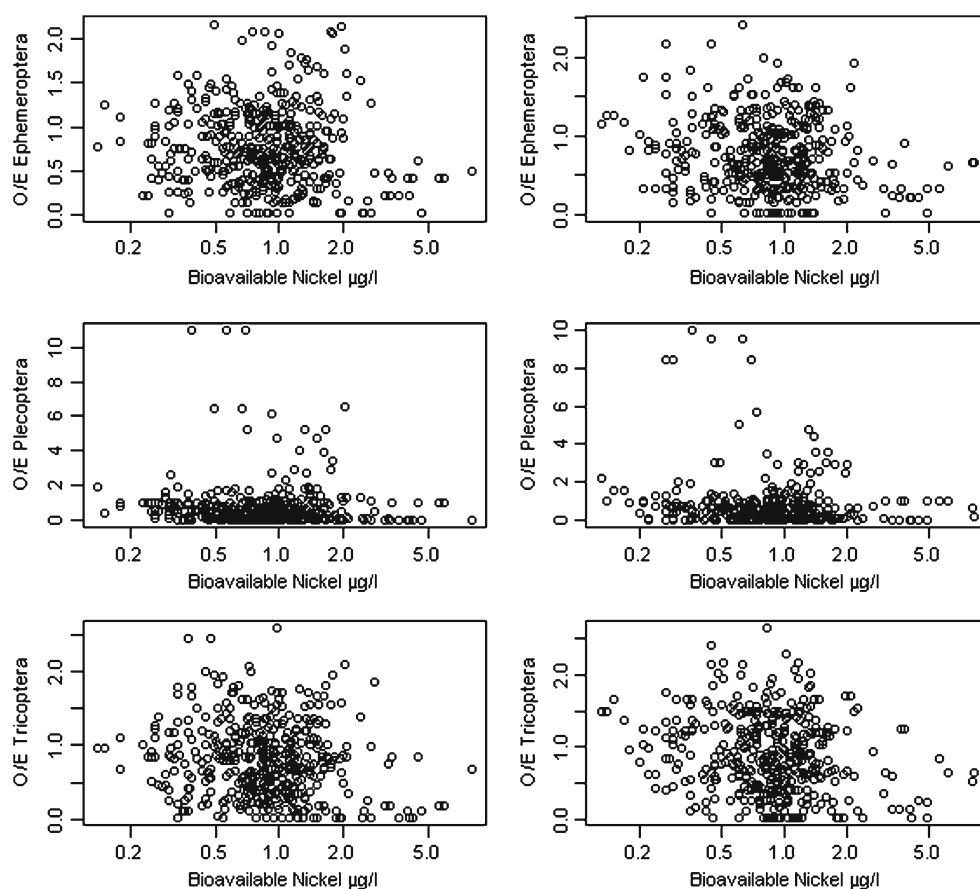
There is much greater variability in O/E values for the individual taxa at low exposure levels, and the quantile regression models which were fitted to identify a decline in ecological quality were not normalised to an O/E value of 1 at low exposure levels. The elevated O/E values for some samples result from the calculation of O/E as the ratio of the reported log abundance score for the site to the predicted log abundance at the site. High O/E values at low exposures will result whenever the observed number of individuals exceeds the predicted number of individuals and the predicted estimate is towards the lower end of the log abundance categories used for recording the number of individuals observed in the samples. This situation occurs because of the low resolution used when recording the number of individuals found in an ecological sample and may also reflect the greater uncertainty inherent in making taxa-specific predictions rather than predictions on the whole-community metrics ASPT and N-TAXA. Abundance measures are not currently used in WFD quality assessment in part due to difficulties in robust predictions relative to reference conditions. RIVPACS III+ incorporates an index based on predictions of abundance termed Q14 (Clarke et al. 2003). The Q14 index is currently considered experimental, and whilst we have undertaken to derive estimates of effect on individual taxa based on RIVPACS predictions, these estimates should be interpreted with some caution.

The threshold of O/E 0.9 is used as an indication of a bioavailable nickel concentration which, if exceeded, could result in a deviation in reference conditions (defined as an O/E ratio of 1) and is broadly comparable to the use of an EC10 from a laboratory ecotoxicity test. Given the lower threshold (0.7) applied for WFD classification based on the whole community, it is possible that a lower threshold of O/E would also be indistinguishable from the reference condition, and in this case, the selected value of O/E 0.9 is considered to represent a precautionary threshold.

Influence of additional stressors on the analysis

One potential limitation of this method is that contaminants whose concentrations co-vary with the contaminant of interest may affect the slope of the dose–response curve or the 90th percentile of values associated with “good” status. This could result in a steeper dose–response curve for the limiting function and a lower threshold. One argument for the use of assessment factors in EQS derivation is that real environmental exposures are to complex mixtures of contaminants, whereas laboratory ecotoxicity tests typically consider exposure only to a single substance. A threshold derived from field data already takes account of other potential pressures. Although the nature of these other pressures may not be clear, the derived threshold accounts for exposure to a

Fig. 7 O/E for Ephemeroptera (*top*), Plecoptera (*middle*), and Tricoptera (*bottom*) abundance as a function of bioavailable nickel concentrations in spring (*left*) and autumn (*right*)



complex mixture of contaminants at low levels, in addition to exposure to the contaminant of interest.

Analyses of field data will invariably include exposures to other contaminants in addition to the contaminant of interest. Whilst the quantile regression approach controls for impacts from other contaminants at sites which have low exposure levels of Ni, any effects which are due to other contaminants which co-vary in their exposures with Ni will be assumed to be due to Ni. In order to assess the possible importance of such effects on this analysis, the exposure dataset was assessed for correlations between dissolved Ni and other reported analytes.

Co-variation was observed between dissolved Ni and copper, zinc, arsenic, chromium, ammonia and phosphate, with r_s values of greater than 0.5 in each case. This assessment did not take account of bioavailability effects on the co-variation of contaminants with Ni, although the most closely co-varying contaminants, dissolved copper (r_s 0.68) and dissolved zinc (r_s 0.57), are also both influenced by the same bioavailability-modifying parameters as Ni. This indicates that sites which have elevated Ni exposures are also likely to have elevated exposures of Cu and Zn. In the case of all three of these metals, DOC is the major influence on bioavailability, and so for sites where Ni bioavailability is high, the bioavailability of copper and zinc will also tend to be high.

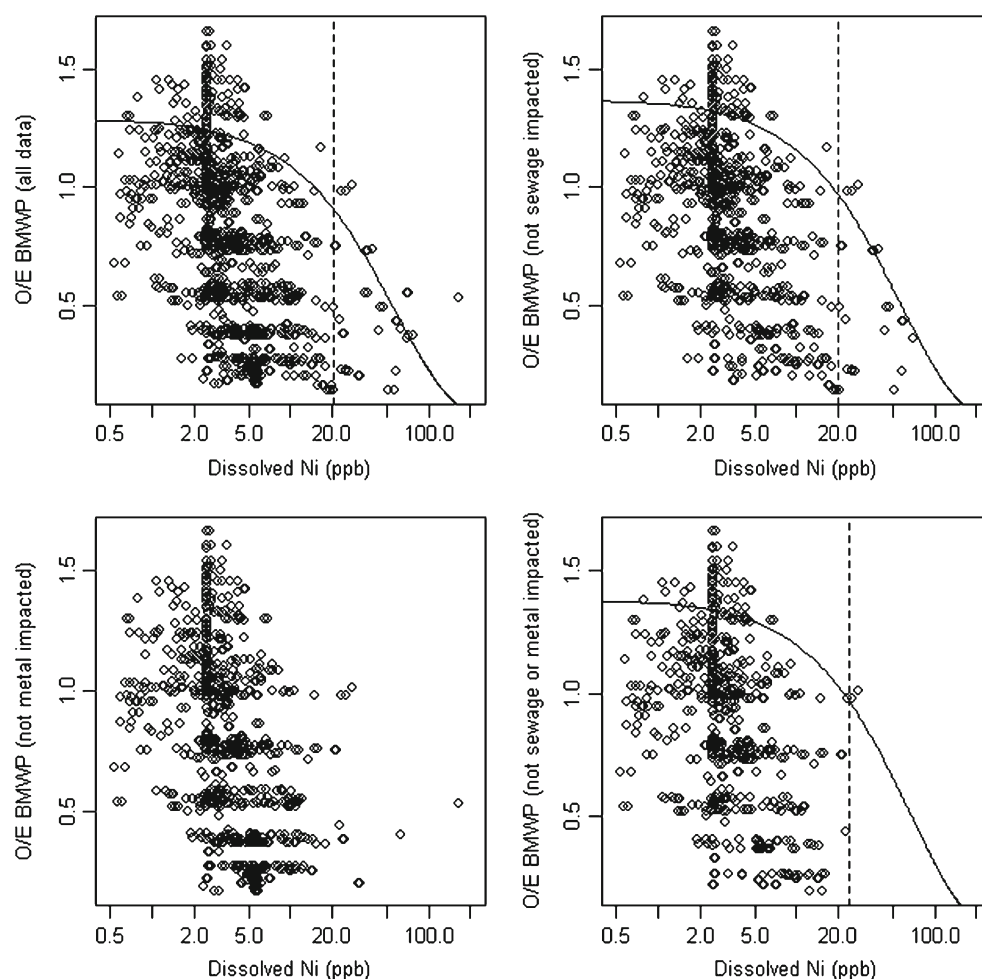
The outcome of this is that the thresholds derived by quantile regression from field data are likely to be lower than they would have been if Ni-only exposures had been considered. Any thresholds derived in this manner will therefore take account of the fact that Ni exposure regimes in real systems are associated with additional pressure from other contaminants, such as copper and zinc. The thresholds that are derived represent an acceptable concentration of Ni in the presence of these additional contaminants.

The possible effect of other metals on the analysis was assessed using data for dissolved Ni and O/E BMWP as an extensive dataset was available. This dataset was screened for sites at which impacts due to sewage contamination were possible (biochemical oxygen demand > 5 mg l⁻¹ or total ammonia > 0.73 mg l⁻¹) and also where elevated levels of trace metals, other than nickel, were identified (dissolved

Table 2 Bioavailable nickel concentrations (µg l⁻¹) at sites achieving good ecological status and the full dataset

Percentile	Good status	All data
90	1.60	2.16
95	1.95	3.39
99	4.97	7.84
Number of samples	674	1,264

Fig. 8 Effect of other pressures on thresholds derived by quantile regression. *Solid lines* indicate the fitted regression model and *dashed lines* indicate the calculated good status threshold



cadmium $>1 \mu\text{g l}^{-1}$, dissolved copper $>16 \mu\text{g l}^{-1}$, dissolved lead $>10 \mu\text{g l}^{-1}$ or dissolved zinc $>30 \mu\text{g l}^{-1}$). These levels were considered to be significantly elevated above typical ambient concentrations. The dataset was assessed in its entirety and also following removal of sewage-impacted sites, metal impacted sites and sites impacted by either sewage or metals (Fig. 8). The status boundary applied here ($\text{O/E}=0.7$) is that used for EQI N-TAXA, and whilst it is not directly related to ecological quality when applied to the EQI BMWP metric, the relative changes between the different data screening approaches allow effective comparisons to be made.

Analysis of the complete data set for O/E BMWP and dissolved Ni resulted in a good status threshold (GST) of $19.6 \mu\text{g l}^{-1}$ dissolved Ni. A similar value for this threshold was also derived from the same dataset with potentially

sewage-impacted sites removed and with sewage- and metal (other than nickel)-impacted sites removed. No statistically significant decline in the maximum achievable ecological quality was observed when potentially metal-impacted sites were removed. The thresholds selected for screening out sites which were potentially impacted by either sewage or other metals were rather arbitrary, although in the case of the metals adverse effects would be anticipated at concentrations around the screening levels used, on the basis of proposed EQS values, except perhaps under low bioavailability conditions.

The GST values derived from each of these datasets are shown in Table 3, along with 95 % confidence intervals. There was little change in the values of the GST for the different analyses, although when both metal- and sewage-

Table 3 Effect of other pressures on the GST ($\mu\text{g l}^{-1}$ dissolved Ni) based on dissolved nickel concentrations, including lower (LCL) and upper (UCL) 95 % confidence limits

Screening applied	GST	LCL	UCL
All data	19.6	19.2	67.2
Not sewage-impacted	19.3	16.6	23.2
Not metal-impacted	No significant decline at the 90th quantile		
Not sewage- or metal-impacted	22.8	9.6	26.6

impacted sites were removed from the analysis the GST was higher than when these sites were included in the analysis.

This also suggests that other contaminants which co-vary in the exposure with Ni can result in thresholds being derived which are lower than they would be for Ni-only exposures. The fact that a statistically significant decline was observed in ecological quality with increasing Ni exposures in three out of the four cases considered here does suggest an effect of Ni on benthic invertebrate communities at elevated exposures. Concentrations of more than $20 \mu\text{g l}^{-1}$ dissolved Ni would tend not to be associated with good ecological status, assuming that the thresholds apply to EQI BMWP. The actual dissolved concentration which causes an effect will depend on the local bioavailability, which was not considered in this analysis.

It is possible that some local environments may be less sensitive than is indicated by this analysis. Local ecosystems in areas which do not tend to naturally support snail populations, perhaps due to factors such as water chemistry, flow, habitat or geology, may be less sensitive to Ni than ecosystems in which snails are particularly important, due to the high sensitivity of snails relative to other groups of benthic invertebrates. Similarly local ecosystems which are less stressed by contamination by other pollutants may also be more resilient to elevated Ni exposures than ecosystems which are already subject to some stress from other contaminants, i.e. where there is a mixture of toxicants.

Conclusions

Estimation of DOC concentrations from dissolved iron concentrations increases uncertainty in HC5 prediction (by an average of 28 %) but does not introduce any systematic bias into the results. Nickel exposure has a limiting effect on benthic macroinvertebrate communities when expressed as EQI N-TAXA. Concentrations below $10.3 \mu\text{g l}^{-1}$ “bioavailable Ni” would not inhibit the ability of sites to achieve good ecological status under the Water Framework Directive.

Nickel exposure does not have a limiting effect on the abundance of Ephemeroptera (mayflies), Plecoptera (stoneflies) or Trichoptera (caddisflies) when expressed relative to a reference condition within the range of “bioavailable Ni” exposures observed within the dataset (up to $8 \mu\text{g l}^{-1}$ bioavailable Ni). Nickel exposure has a limiting effect on snail abundance, although snail abundance is only slightly affected relative to a reference condition at a “bioavailable Ni” concentration of $\leq 3.9 \mu\text{g l}^{-1}$.

Dissolved Ni exposures co-vary with copper, zinc, arsenic, chromium, ammonia and phosphate with Spearman's correlation coefficients of greater than 0.5. Thresholds derived in the presence of these contaminants will be lower than thresholds derived on the basis of Ni-only exposures

and will be protective of the mixed toxicant exposures that occur in the environment.

Understanding the importance of effects on snail abundance on community quality under different kinds of conditions may be able to provide further information on the ecological relevance of the snail abundance thresholds derived in this study.

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